THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–11) | Class 7 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** an acid	Acid.
2	**(d)** glow dimly	4.5 V is less than the rated 6 V.
3	**(a)** magnesium	Basic oxide, so metal.
4	**(d)** a reversible chemical change	Hydrated and anhydrous forms.
5	**(b)** growth rates vary and the cousin may have started his spurt	Individuals grow differently.
6	**(a)** A is a better conductor	Faster fall means faster conduction.
7	**(c)** 20 m/s	(150+250)/20.
8	**(b)** fat-rich food	Bile aids fat digestion.
9	**(b)** the covered half remains brown with iodine	Light needed.
10	**(c)** returns along the same path	For 90°, reversed.
11	**(c)** more extreme	Larger variation.
12	**(d)** 40 mL	Proportional.
13	**(b)** test with a tester; add salt to water to make it conduct	Dissolved salts supply ions.
14	**(d)** magnesium	Magnesium is highly reactive.
15	**(d)** formation of a new substance	New substance.
16	**(b)** larynx is growing and adjusting	Settles in time.
17	**(a)** Both A and R are true and R is the correct explanation of A.	Turmeric turns red (reddish brown) with bases, and soap solution is a base, so R explains A.
18	**(a)** Both A and R are true and R is the correct explanation of A.	Current flows only in a complete (closed) circuit, so R explains why the bulb glows.
19	**(a)** Both A and R are true and R is the correct explanation of A.	Malleability is the property of being beaten into thin sheets, so R explains A.
20	**(a)** Both A and R are true and R is the correct explanation of A.	Only the state changes from solid to liquid, so R explains why it is physical.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Lemon juice is acidic (1); It neutralises the base so turmeric turns back to yellow (1).	1+1
22	Tungsten has a very high melting point (1); Copper would melt before glowing (1).	1+1
23	$Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$ (1); Zinc sulphate and hydrogen (1).	1+1
24	Iron combines with oxygen and water (1); Rust has extra mass (1).	1+1
25	Hormones carry messages through blood (1); They control growth and mood (1).	1+1
26	Heating from the top gives little convection (1); From the bottom, hot liquid rises and cooler sinks (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Ratio 10:20 (1); 60 is three times (1); 30 mL acid (1).	1+1+1
28	6 x 1.5 = 9 V (1); Much higher than 3 V (1); Bulb may fuse (1).	1+1+1
29	Metal (1); Magnesium (1); Basic oxide (1).	1+1+1
30	A: no rust (no moisture) (1); B: no rust (no air) (1); C: rusts (air and water) (1).	1+1+1
31	Growth hormone lacking (1); Poor growth (1); Other glands underactive (1).	1+1+1
	OR: Thermocol best insulator (1); Ceramic next (1); Metal loses most heat (1).	1+1+1

32	Speed = 120/12 = 10 m/s (1); Distance in 30 s = 300 m (1); Platform = 300 - 120 = 180 m (1).	1+1+1
	OR: Fat digestion poor (1); Fewer enzymes (1); Special diet needed (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) Hydrochloric acid (1). (ii) It is a base (an antacid) (1). (iii) Neutralisation (1). (iv) Lemon juice is acidic (it contains citric acid) (1), so it would add more acid to the stomach and make the burning worse instead of neutralising the extra acid (1).	1+1+1+2
34	(i) $1.5 + 1.5 = 3$ V (1). (ii) The switch (1). (iii) The filament (1). (iv) The filament may be broken (fused) or there may be a loose connection (1); in both cases the circuit is not complete, so no current flows through the filament (1).	1+1+1+2
35	Colour and lustre (1); Conduction test (1); Magnet for iron (1); Burning gives acidic or basic oxide (1); Reaction with acid (1).	1+1+1+1+1
36	Set-ups: dry, no air, tap water, salt water (1); Same nails and period (1); Observe daily (1); Compare rust amounts (1); Salt water most, oiled none (1).	1+1+1+1+1
	OR: Poor energy and concentration (1); Disturbed sleep (1); Weight and mood problems (1); Plan: breakfast, sleep time, screen limits (1); Monitor progress (1).	1+1+1+1+1
37	Same volume and initial temperature (1); Wrap each beaker (1); Measure every 5 min (1); Same room conditions (1); Wool keeps heat longest (1).	1+1+1+1+1
	OR: 0-10 s: 2 m/s (1); 10-15 s: rest (1); 15-20 s: 6 m/s (1); Average 50/20 = 2.5 m/s (1); Graph has a flat part (1).	1+1+1+1+1
38	Rest: count breaths per minute (1); After walking (1); After running (1); Table of results (1); Rate rises with activity (1).	1+1+1+1+1
	OR: Vary lamp distance (1); Count bubbles per minute (1); Keep temperature and CO2 same (1); Table and graph (1); Rate rises with light (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–11) | Class 7 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** umbra	Complete shadow.
2	**(b)** more frequent leap years	Larger quarter.
3	**(d)** acidic	Acid rain.
4	**(c)** a conductor	Current passes through it.
5	**(b)** zinc	Acid reacts.
6	**(a)** store in dry place or oil them	Keeps moisture away.
7	**(b)** a rapid rise during the growth spurt	Sudden increase.
8	**(a)** sea breeze	Fluid motion.
9	**(d)** 1.82 s	Mean 18.2 / 10.
10	**(d)** to provide a large surface area for absorption	Large area.
11	**(a)** nitrogen-fixing bacteria	Rhizobium.
12	**(c)** virtual image position follows the law of reflection	Geometric consequence.
13	**(d)** the Earth always has a shadow but the Moon enters it only during eclipses	Shadow is always there.
14	**(d)** an acid	An acid neutralises the base.
15	**(b)** glow very bright and may fuse	Exceeds rating.
16	**(b)** sulphur	Non-metal.
17	**(a)** Both A and R are true and R is the correct explanation of A.	Growth spurts and other changes of the teenage years are brought about by hormones.
18	**(a)** Both A and R are true and R is the correct explanation of A.	R describes the result of the flow from hot to cold, supporting A.
19	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the relation between minute and second does not explain why the second is the SI unit.
20	**(a)** Both A and R are true and R is the correct explanation of A.	The enzyme in saliva starts acting on starch in the mouth, so R explains A.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Distance 500 m in 30 s (1); 500/30 = 16.67 m/s, i.e. 60 km/h (1).	1+1
22	Respiration releases energy as heat (1); Metabolism increases after a meal (1).	1+1
23	The plant uses CO2 produced and gives oxygen (1); The candle can burn longer (1).	1+1
24	Angle of incidence is 0° (1); The ray retraces its path (1).	1+1
25	Moon reflects only a small fraction (1); Light is weak (1).	1+1
26	Use litmus or turmeric (1); Different colours appear (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Hydrilla in water under funnel (1); Test tube collects gas (1); Control in dark (1).	1+1+1
28	i = r = 55° (1); Angle between rays 110° (1); Angle with mirror 35° (1).	1+1+1
29	15° in 1 hour (1); 1° in 4 minutes (1); 15° takes 1 hour (1).	1+1+1
30	Some are more sensitive (1); Colour changes differ (1); Choose according to need (1).	1+1+1
31	They need a large current (1); Thin wiring may overheat (1); Risk of fire (1).	1+1+1
	OR: $MgO + H_2O \longrightarrow Mg(OH)_2$ (1); $SO_2 + H_2O \longrightarrow H_2SO_3$ (1); Basic and acidic (1).	1+1+1

32	Salt and sand do not react (1); Iron and sulphur react (1); New compound formed (1).	1+1+1
	OR: Avoid excess sweets (1); Carry medicine (1); Eat at regular times (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) Metals are good conductors of heat (1). (ii) Plastic is a poor conductor of heat, so the handle does not become hot (1). (iii) Malleability (1). (iv) Ductility (1); copper is also a good conductor of electricity (1).	1+1+1+2
34	(i) Crystallisation (1). (ii) Physical change (1). (iii) More copper sulphate dissolves in hot water, so the solution becomes saturated and crystals form on cooling (1). (iv) Cool the solution slowly and keep it undisturbed (1), so the particles have time to arrange themselves and grow into larger crystals (1).	1+1+1+2
35	Some crops prefer acidic soil (1); Others basic (1); Lime reduces acidity (1); Manure improves soil (1); Test regularly (1).	1+1+1+1+1
36	Marks show quality (1); Safer products (1); Fuse rating matches use (1); Do not use higher rating (1); Check before buying (1).	1+1+1+1+1
	OR: Burn Mg ribbon (1); Dissolve ash in water (1); Burn sulphur and dissolve gas (1); Test both with litmus (1); Compare (1).	1+1+1+1+1
37	Burn a candle in a closed jar (1); It goes out (1); Add lime water (1); It turns milky (1); Conclude (1).	1+1+1+1+1
	OR: Respect for others (1); Say no to unwanted behaviour (1); Report abuse (1); Trusted adults (1); Laws protect (1).	1+1+1+1+1
38	Black in sun highest (1); White in sun lower (1); Both in shade low (1); Differences show absorption (1); Conclude (1).	1+1+1+1+1
	OR: Distance 80 km in 2 h (1); Average 40 km/h (1); Speedometer shows instantaneous (1); Values differ (1); Both useful (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER

ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–11) | Class 7 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** digestion of all three food types	Pancreatic juice is important.
2	**(a)** chlorophyll is needed	Pigment.
3	**(d)** increases by 15°	Twice the turn of the mirror (10 x 2).
4	**(c)** 12 hours	Half.
5	**(d)** 30 mL	Doubling.
6	**(a)** tap water has dissolved salts	Dissolved ions conduct.
7	**(c)** Fe	Iron.
8	**(a)** lime water test	Test gas.
9	**(d)** menstruation	Ovary hormones needed.
10	**(a)** nearly stops	Temperatures equalise.
11	**(c)** 360 km	Gap 120 km, closing 30 km/h in 4 h.
12	**(b)** 8000 mL	16 x 500.
13	**(a)** chemical energy to usable energy	Released.
14	**(a)** shadow	Dark.
15	**(a)** the Earth moves around the Sun meanwhile	Extra two days to realign.
16	**(c)** it is a base used to relieve acidity	Carbonates react with acids.
17	**(a)** Both A and R are true and R is the correct explanation of A.	The green pigment traps light energy, which explains why chlorophyll is needed.
18	**(a)** Both A and R are true and R is the correct explanation of A.	Because light cannot bend around the opaque object, a dark region (shadow) forms behind it.
19	**(a)** Both A and R are true and R is the correct explanation of A.	Rotation turns each part of the Earth to and away from the Sun every 24 hours.
20	**(d)** A is false but R is true.	Acids turn blue litmus red and do not change red litmus; so A is false, but R is true.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Metals are good conductors (1); They carry current to the appliance (1).	1+1
22	They dissolve in rain to form acids (1); Such as SO_2 giving sulphurous acid (1).	1+1
23	Burning forms ash and gases (1); Tearing only changes size and shape (1).	1+1
24	Testosterone promotes muscle growth (1); Greater rise in boys (1).	1+1
25	Evaporation takes heat (1); Keeps water cool (1).	1+1
26	Time spent at each speed differs (1); Average speed = total distance/total time (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Steel conducts (1); Plastic does not (1); Steel handle becomes hot (1).	1+1+1
28	Time = 3 h (1); Distance = speed x time (1); 40 x 3 = 120 km (1).	1+1+1
29	Move to site (1); Engulf germs (1); Defence (1).	1+1+1
30	Light is available at the top (1); Competition for sunlight (1); More photosynthesis (1).	1+1+1
31	Fix a stick vertically (1); Mark shadow lengths (1); Shortest shadow is at noon (1).	1+1+1
	OR: Earth's shadow is bigger (1); Moon's shadow small (1); Speeds (1).	1+1+1
32	$H_2SO_4 + 2\,KOH \longrightarrow K_2SO_4 + 2\,H_2O$ (1); Acid and base (1); Salt and water (1).	1+1+1

	OR: Switch off the supply (1); Dry hands (1); Use proper bulb (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) Puberty (adolescence) (1). (ii) Testosterone (1). (iii) The testes (1). (iv) The voice box (larynx) grows larger at puberty (1), so the voice becomes deeper; in boys the enlarged voice box can be seen as the Adam's apple (1).	1+1+1+2
34	(i) Conduction (1). (ii) Wood is a poor conductor (insulator) of heat (1). (iii) Convection (1). (iv) Water at the bottom gets heated, becomes lighter and rises (1); cooler, heavier water from the top moves down and a convection current heats all the water (1).	1+1+1+2
35	Pinhole real inverted (1); Mirror virtual erect (1); Pinhole needs a screen (1); Mirror image behind (1); Both use straight-line light (1).	1+1+1+1+1
36	Regular motions (1); Records of old eclipses (1); Calculations (1); Computer models (1); Accurate predictions (1).	1+1+1+1+1
	OR: Soil treatment (1); Antacids (1); Acid rain (1); Treating wastes (1); Principle: acid + base gives salt + water (1).	1+1+1+1+1
37	Switch off unused devices (1); Use LED bulbs (1); Use daylight (1); Do not overload (1); Service appliances (1).	1+1+1+1+1
	OR: $2\,Mg + O_2 \longrightarrow 2\,MgO$ (1); $Mg + 2\,HCl \longrightarrow MgCl_2 + H_2$ (1); $Mg + CuSO_4 \longrightarrow MgSO_4 + Cu$ (1); Mg is reactive (1); Products observed (1).	1+1+1+1+1
38	LPG burns: chemical (1); Milk warms: physical (1); Milk boils: physical (1); Milk burns at bottom: chemical (1); Black deposit is carbon (1).	1+1+1+1+1
	OR: Nerves: fast signals (1); Hormones: slower, lasting (1); Adrenaline for stress (1); Sex hormones for puberty (1); Both coordinate (1).	1+1+1+1+1

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